Searching for Multiple Attributes

Using indices for multiple attributes

What if queries like the following are common?

```
SELECT …
FROM Employee
WHERE job_code = 2 AND perf_rating = 5;
```

Strategy 1 – Index one attribute

```
CREATE INDEX idx_job_code ON Employee (job_code);

SELECT …
FROM Employee
WHERE job_code = 2 AND perf_rating = 5;
```

Internal strategy:

1. Use index to find Employees with job_code = 2.
2. Linear search of those to check perf_rating = 5.

Strategy 2 – Index each attribute

```
CREATE INDEX idx_job_code ON Employee (job_code);
CREATE INDEX idx_perf_rating ON Employee (perf_rating);

SELECT …
FROM Employee
WHERE job_code = 2 AND perf_rating = 5;
```

Internal strategy – chooses between

- Use job_code index, then linear search on perf_rating.
- Use perf_rating index, then linear search on job_code.
- Use both indices, then intersect resulting sets of pointers.

Strategy 3 – Index attribute set

```
CREATE INDEX idx_job_perf ON Employee (job_code, perf_rating);

SELECT …
FROM Employee
WHERE job_code = 2 AND perf_rating = 5;
```

Order:

1,1
1,2
1,3
1,4
1,5
2,1
2,2
2,3
2,4
2,5

Attribute sets ordered lexicographically:

Needs an ordered index.

(jc1, pr1) < (jc2, pr2) iff either

- jc1 < jc2
- jc1 = jc2 and pr1 < pr2

Note that this prioritizes job_code over perf_rating!

Strategy 3 – Index attribute set

```
CREATE INDEX idx_job_perf
ON Employee (job_code, perf_rating);
```

Order:
1,1
1,2
1,3
1,4
1,5
2,1
2,2
2,3
2,4
2,5

Efficient

```
SELECT … WHERE job_code = 2 AND perf_rating = 5;
SELECT … WHERE job_code = 2 AND perf_rating < 5;
```

Inefficient

```
SELECT … WHERE job_code < 2 AND perf_rating = 5;
```

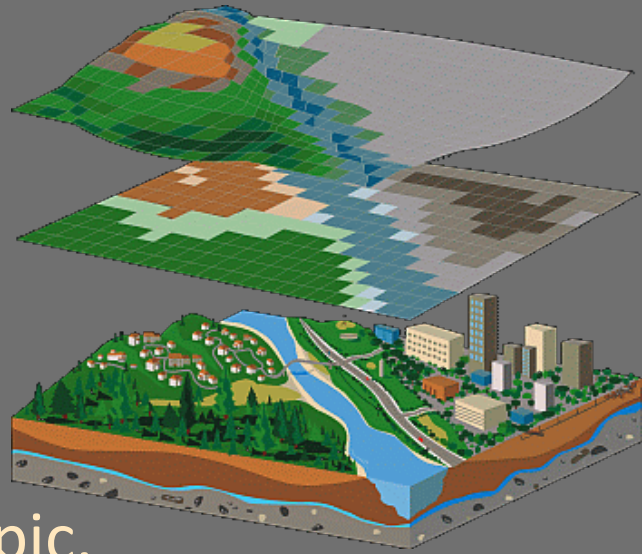
Left-most index columns should have the most selective conditionals (typically equality).

Inefficient

```
SELECT … WHERE job_code = 2 OR  perf_rating = 5;
```

Strategy 4 – Grid indexing

- n attributes viewed as being in n-dimensional grid
  - Mainly used for spatial data



- Numerous implementations
  - Grid file, K-D tree, R-tree, …

- GIS DBs are a separate, specialized topic.
  - Many RDBMS's have a GIS-specialized component.

Strategy 5 – Bitmap indices

Coming …